SCHOOL OF COMPUTING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
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Course code: 10214AD701 **Course Name: Major Project**

TTS NO	Supervisor Name	VTU NO.	Student Name1	VTU NO.	Student Name 2	VTU NO.	Student Name 2
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Title of the Project: **Hybrid Photovoltaic and Ducted Energy Integration for Dynamic EV Charging**

Problem Statement

Electric vehicles face major challenges such as dependence on stationary charging stations, long charging time, and inconsistent energy availability. Existing systems rely on single energy sources and lack real-time energy management, making them inefficient and unreliable. Therefore, there is a need for a hybrid renewable energy-based dynamic charging system that enables continuous charging during vehicle motion while ensuring efficient energy utilization and battery safety.

Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKs (wherever applicable)	Justification (as applicable)
WP1	Depth of Knowledge	Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8	WK3, WK4, WK5, WK6, WK8	The project requires strong engineering knowledge in renewable energy systems, power electronics, embedded systems, and control engineering. It involves designing hybrid energy models (solar PV + wind), implementing DC–DC converters, and developing a Dynamic Energy Management System

				(DEMS), which demands both theoretical understanding and practical expertise.
WP2	Range of conflicting requirements	Involve wide-ranging and/or conflicting technical and non-technical issues	WK4, WK5, WK6, WK8	The system must balance multiple conflicting requirements such as maximizing energy generation vs. minimizing system losses, ensuring battery safety vs. achieving faster charging, and maintaining efficiency under varying environmental conditions. It also considers sustainability, cost, and real-time performance constraints.
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	There is no direct or standard solution for dynamic EV charging using hybrid renewable sources. The project requires analytical modeling of solar and wind energy generation, real-time decision-making algorithms, and performance optimization under varying vehicle speeds and environmental conditions.
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	The problem involves relatively new and evolving concepts such as dynamic charging during vehicle motion and hybrid energy harvesting. These are not commonly addressed in traditional EV systems, making the problem novel and less familiar.
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice	WK6, WK8	Existing EV charging standards mainly focus on stationary charging systems and do not fully cover hybrid dynamic charging using multiple renewable sources. Hence, the project relies on

				research-based approaches and engineering judgment rather than predefined standards.
WP6	Stakeholder involvement and conflicting requirements	Involve collaboration across disciplines and diverse stakeholders	WK5, WK6, WK7	The system involves multiple stakeholders including EV users, manufacturers, energy providers, and environmental regulators. Each has different expectations such as cost efficiency, reliability, safety, and sustainability, which must be addressed within the system design.
WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	The project consists of multiple interdependent subsystems including solar PV modules, wind turbine systems, power converters, battery storage, sensors, and the DEMS controller. The performance of the overall system depends on the coordinated functioning of all these components, requiring a systems engineering approach.

Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project uses diverse resources including solar PV panels, ducted wind turbine, Arduino Nano, sensors, battery systems, and power electronics, along with Arduino IDE for DEMS implementation.
EA2	Level of Interactions	Optimization of complex interactions	It involves interaction between energy generation, power conversion, battery charging, and DEMS control to ensure efficient and stable operation under varying conditions.
EA3	Innovation	Creative and research-based solutions	The project introduces a hybrid solar–wind dynamic EV charging system with real-time DEMS control, enabling charging during vehicle motion.
EA4	Consequences to Society and Environment	Significant societal impact	It promotes sustainable transportation by reducing dependence on charging stations and using renewable energy sources.

EA No.	Attribute	Characteristics	Justification based on the Project
EA5	Familiarity	Extends beyond prior experience	The project applies engineering principles to a novel problem of dynamic hybrid EV charging beyond conventional systems.

Sustainable Development Goals (SDG) Mapping

SDG No.	SDG Title	Relevance to the Project
SDG 7	Affordable and Clean Energy	Uses solar and wind energy for EV charging, promoting clean and sustainable power generation.
SDG 9	Industry, Innovation and Infrastructure	Develops an innovative hybrid EV charging system, improving sustainable transportation infrastructure.
SDG 11	Sustainable Cities and Communities	Supports eco-friendly mobility and reduces urban pollution through continuous EV charging.
SDG 13	Climate Action	Reduces carbon emissions by integrating renewable energy into EV charging systems.

Overall Justification: The project satisfies Complex Engineering Problem and Activity characteristics by involving multidisciplinary knowledge in renewable energy, power electronics, and embedded systems. It addresses real-world challenges such as dynamic EV charging, energy variability, and battery safety using a hybrid solar–wind system with DEMS. The project involves system design, analysis, and contributes to sustainable transportation.

Depth of Knowledge – Required Knowledge Levels (K3–K8)

The problem addressed in this project cannot be resolved without in-depth engineering knowledge spanning the following **Knowledge Profile levels**:

K3 – Engineering Fundamentals

- Understanding of electrical circuits, solar PV, and wind energy principles.
- Basics of battery charging and energy conversion systems.

K4 – Specialist Knowledge

- Knowledge of hybrid renewable energy systems combining solar and wind.
- Understanding of power electronics such as rectifiers and DC–DC converters.

K5 – Engineering Design

- Design of hybrid EV charging system integrating solar, wind, and battery.
- Development of DEMS for real-time monitoring of voltage, temperature, and SoC.

K6 – Engineering Practice

- Implementation using Arduino Nano, sensors, and hardware components.
- Testing and validation under varying environmental and operating conditions.

K7 – Engineering Comprehension

- Analysis of system performance such as voltage stability and efficiency.
- Interpretation of results for safe and optimized battery charging.

K8 – Research Literature and Emerging Technologies

- Use of recent research in EV charging and hybrid energy systems.
- Application of research insights to develop dynamic charging solutions.

Signature